CT-GAN: A Cross-Modal Transformer GAN for Superior Multi-Modal MRI-Based Alzheimer's Disease Detection

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Abstract-

This research introduces CT-GAN, an innovative Cross-Modal Transformer employing Generative Adversarial Networks for the diagnosis of Alzheimer's Disease (AD) utilizing multi-modal neuroimaging data. CT-GAN accurately delineates the spatial and functional intricacies of the brain by integrating functional MRI (fMRI) and Diffusion Tensor Imaging (DTI) with a bi-attention mechanism. CTGAN consistently outperformed state-of-the-art generative models—DCGAN, WGAN, CGAN, Pix2Pix, CycleGAN, and VAE-GAN—on all the important criteria, with the best accuracy (90%), precision (89%), recall (91%), F1-score (90%), and an amazing AUC of 0.95. They show that CT-GAN can make high-quality fake data without hurting its ability to classify things well, with few false positives and negatives. It works well in high-risk medical settings because it has a better balance between precision and recall. The results show that professional designs like CT-GAN are very important for working with complicated multi-modal data. They also show that CT-GAN could be a useful tool for analyzing and diagnosing medical images, not just for AD diagnosis.

Keywords: *CT-GAN, Alzheimer's Disease detection, multi-modal data fusion, functional MRI, Diffusion Tensor Imaging, bi-attention mechanism, generative models, image generation, precision, recall, accuracy, F1-score, Area Under the Curve (AUC).*

I. INTRODUCTION

A progressive neurological disease that affects millions of individuals and their families worldwide, Alzheimer's disease (AD) is characterized by memory loss, cognitive impairment, decline in cognitive abilities, and changes in behavior. 5.1 million Americans have been found to suffer from the disorder at present, and its prevalence will continue to rise with the population of the world's aging population. Alzheimer's disease must be found in its early stage only so that doctors can control and treat it properly. Even though a complete cure cannot be claimed at the moment, early diagnosis has the potential to stop disease processes with medication regimens, life- (Zhang et al., 2023) style modification, and therapy modalities. Early detection enables individuals and carers to plan their future, i.e., long-term care, legal and financial affairs, without it influencing the autonomy and quality of life of the individuals. Magnetic resonance imaging (MRI) has become an essential imaging modality for depicting structural changes in the brain related to dementia, like cortical flattening and reduction of the hippocampus. Getting good MRI data for training diagnostic models is actually very difficult because of privacy problems, limited patient access, and money issues. These challenges definitely make it hard to build effective medical AI systems. Further, these limits actually reduce how well machine learning systems work across different groups of people. They definitely make it harder to find Alzheimer's disease in its early stages. Scientists are using generative artificial intelligence (AI) techniques to data augment and circumvent these limitations. The promise of generative AI is to transform Alzheimer's diagnosis by generating simulated MRI scans that are remarkably similar to actual patient scans, thereby increasing the availability and variety of datasets and maintaining the confidentiality of the patients. Variational auto encoders (VAEs) and Generative Adversarial Networks (GANs) are just a few instances of novel AI models. have revolutionary promise in medical imaging. High-quality synthesized MRI images can be produced through generative AI algorithms. augmenting current datasets, and global optimization of diagnostic algorithm training, in Alzheimer's disease, where global minor alterations in brain structure are crucial for early detection, generative AI enhances diagnostic accuracy by providing global variability in presentations of data over phases and anatomical variations. In addition, the models can denoise data and improve image quality, reducing the difficulty of detecting early illness biomarkers by clinicians and machines. The ability of generative AI to mimic disease progression provides new avenues for the training of diagnostic models such that researchers and clinicians can observe the temporal progression of Alzheimer's from its beginning to late stages. Apart from data augmentation, generative AI also addresses major practical and ethical challenges of Alzheimer's research. AI-generated synthetic datasets don't have clear patient identities, so they can

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be shared and worked on by many researchers while still following strict privacy rules. Making data available to everyone leads to more innovation, faster development of diagnostic tools, and ensures that better ways to find Alzheimer's are available in all healthcare settings. Also, generative AI makes it possible to create and use advanced diagnostic models in places with few resources by reducing the need for expensive and time-consuming data collection methods. Early diagnosis is very important. Early diagnosis lets patients get newer treatments, join clinical trials, and take steps to prevent cognitive decline. Early interventions have been demonstrated to alleviate the economic burden on the healthcare system by prolonging the independence of affected individuals and postponing the necessity for intensive care. Using generative AI for Alzheimer's diagnosis itself brings major improvements in detection rates. Further, this technology helps understand the disease mechanisms and causes better. MRI technology with AI actually helps solve the big problems that old diagnostic methods have when finding Alzheimer's disease. This combination definitely works better than traditional tools for detecting the condition. Basically, generative AI gives better quality data and makes diagnosis faster and more accurate, which helps patients get the same improved treatment for brain diseases. This study looks at how generative AI can help improve data for detecting Alzheimer's using MRI scans. We are seeing that this technology can only transform early diagnosis and help doctors manage patients better.

(Li et al., 2024)

II. Related Work

III. Here, we introduce the current research in medical image acquisition and object detection methods. The preceding studies' findings and observations are summarized and provided, well summarizing the research gaps that were found.

Study	Modality	Dataset	Methodology	Advantages	Disadvantages
[1]	MRI	ADNI	Utilized Latent Diffusion Models and Convolutional Neural Networks to improve AD detection.	Demonstrated enhanced detection accuracy by combining generative models with CNNs.	Requires extensive computational resources for training complex models.
[2]	MRI	Various	Reviewed AI applications in brain MRI analysis for AD, focusing on deep learning approaches.	Provided a comprehensive evaluation of AI methodologies in MRI-based AD analysis.	Lacked experimental validation of discussed methods.
[3]	MRI	Multiple cohorts	Systematically reviewed AI applications in neuroimaging for dementia diagnosis and prognosis.	Offered insights into the integration of AI in clinical settings for dementia care.	Did not focus exclusively on MRI modalities.
[4]	MRI	Bibliometric data	Conducted a bibliometric analysis on AI applications in early AD diagnosis and monitoring.	Identified research trends and gaps in AI-based AD diagnostics.	Relied on bibliometric data without experimental analysis.
[5]	MRI and Genomics	ADNI	Developed an interpretable generative multimodal framework combining neuroimaging	Enhanced understanding of AD by integrating multimodal data with interpretability.	Complexity in integrating diverse data types and ensuring interpretability.

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			and genomics for AD decoding.		

IV. METHODOLOGY

Data Collection

The GAN training data comes from the Kaggle dataset "images-oasis" itself. This dataset further provides the necessary images for model training. It can be downloaded using the Kaggle hub library. After downloading, the dataset itself is copied to the correct working directory for further processing. All images are resized to 128 × 128 pixels because deep learning models need same size inputs for proper training. GANs themselves require this uniformity to further improve their training speed. This resizing surely makes all data uniform across the dataset, which helps both generator and discriminator networks learn patterns more easily. Moreover, this standardization improves the overall training process for both networks. Moreover, a custom preprocessing function is used to further improve the quality of training data itself. This function surely crops and resizes bounding boxes in images when they are present. Moreover, it handles the main task of adjusting box dimensions for image processing. As per the function's purpose, it clearly defines objects of interest by adding padding around their bounding boxes in the surrounding areas. It is observed that quality improves only when over-cropping is stopped. This method prevents too much cutting of images. This operation surely helps when the dataset has important objects that need special attention during training. Moreover, it makes the model focus better on these key elements. After preprocessing, the dataset is surely divided into different groups for training and validation purposes. Moreover, this division helps in better model development and testing. TensorFlow's image_dataset_from_directory function actually makes loading image datasets very simple. This function definitely helps organize images from folders into training datasets without complex coding.

This is an automated naming process according to the directory structure of the images, shuffles the data, and transforms them into TensorFlow datasets for pipelined processing. The validation dataset is utilized in optimizing GAN, while the learning dataset monitors the performance and generalization ability of the model. This is a methodical method to achieving a well-prepared dataset, which is essential in training an effective and stable GAN model.

Hyperparameter Configuration

To get the Cross-Modal Transformer GAN (CT-GAN) architecture's Generative Adversarial Network (GAN) to train well, some of the most important hyperparameters are chosen and carefully adjusted to find the right balance between stability and performance. The training lasts for five epochs, but it can be made better if the quality of the images generated gets better. Batch size is kept at 128 to enable effective utilization of GPU memory and smooth training processes. Learning rate is kept at 2×10^{-4} to enable gradual generator and discriminator weight updates, without causing issues such as mode collapse or vanishing gradients. The noise vector size is fixed at 64, which gives the generator a latent space representation that lets it make a wide range of realistic samples. The GAN architecture has multiple layers in both the generator and the discriminator. The depth and width of these layers are changed to make them more expressive and faster to compute. The generator builds noise vectors into real images one at a time. The discriminator, on the other hand, correctly tells the difference between real and generated examples and helps the generator get better. The bi-attention mechanism makes the process easier by combining resting-state fMRI functional information with structural information from Diffusion Tensor Imaging. This helps find Alzheimer's Disease (AD) more easily. To make training more robust and generalizable, adaptive data augmentation is included. The method subject's images to controlled distortion during training so that the model is able to become more robust to variations. The augmentation consists of a maximum translation of 0.125 (moving images on the x and y axes), a maximum rotation of 0.125 (imposing small angular differences), and a maximum zoom of 0.25 (scaling images within a predetermined interval). These augmentations avoid the discriminator overfitting to certain features, making it learn more generalizable representations. In addition, adaptive augmentation is steered by a target discriminator accuracy of 85%. This implies that augmentation power is dynamically changed according to the accuracy of the discriminator—if the discriminator becomes overly confident (over 85%), more powerful augmentations are used to add more diversity to the training data. When accuracy falls too low, augmentation power is decreased in order to preserve training stability. This method provides an active balance between discriminator and generator, resulting in more effective and stable training, which is paramount for improving the capability of CT-GAN to integrate multimodal neuroimaging data for AD detection.

Model Design

In this cross-modal Transformer GAN model design, the Generator of the CT-GAN employs transposed convolutional layers, or deconvolution layers, to iteratively expand an array of noise to the size of a 128×128×3 image. It starts from a low-dimensional latent space, restoring it to a higher one. Every sampling layer increases the size of the image and improves its structure. After every convolutional layer, sequential normalization is applied to normalize the activations, stabilize training, and accelerate resolution. To facilitate the framework to learn complex patterns and correlations, non-linearity is introduced throughout the system through the implementation of ReLU function activation. The last layer of the generator applies the sigmoid activation function, which normalizes the output pixel values to the range between 0 and 1, thus making the generated image have normalized pixel values for visualization or additional processing. The Discriminator in CT-GAN consists of a sequence of convolutional layers with strides 2, which downsamples the spatial dimensions of the input image. This helps the model focus on high-level features and ignore other unnecessary details. Leaky ReLU activation functions take the place of regular ReLU and give negative

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inputs a small, non-zero gradient. This keeps the model from "dying" while it learns and makes it better at learning. The last layer of the discriminator is a dense layer that outputs a score between 0 and 1 that tells you how likely it is that an image is real or fake. So, the discriminator is a binary classifier that sorts real and fake photos in the GAN system. This structure is important because it helps the CT-GAN effectively combine multimodal neuroimaging data and improve the detection of AD.

Adaptive Augmenter

The Adaptive Augmenter is a flexible way to get the most out of the augmentation process when training a Generative Adversarial Network (GAN). The Adaptive Augmenter changes the chances of doing certain image augmentations, like random horizontal flip, translations, rotation, and zoom, based on how well the discriminator is doing at the time. The augmenter usually changes the chance of each augmentation at each training step by keeping an eye on how accurate the discriminator is in real time. The goal is to keep the target discriminator's performance at a level where the model won't fit too closely to certain patterns or features. If the discriminator is picking up too quickly and only distinguishing real from fake images, then the augmenter can increase the likelihood of applying augmentations such that the task is still difficult enough and keep the model from learning the basic features by heart. Conversely, if the discriminator is having a tough time, the probability of augmentation is decreased, allowing the model to focus more on recognizing genuine patterns rather than handling too many distortions. This self-adjusting mechanism provides a balanced and productive training process, allowing the model to generalize well and enhancing the overall performance of both the generator and discriminator.

Training Process

The training procedure for the Cross-Modal Transformer GAN (CT-GAN) focuses on balanced and iterative optimization policy between the Generator and Discriminator. The Generator Loss is calculated Utilizing binary cross-entropy, a popular option in GANs, the engine aims to trick its discriminator by producing visuals. that appear indistinguishable from real images. This loss function compares the one who discriminated against the predicted label (for the pictures that were produced) to a real label (set to 1, indicating the image is real). The generator's objective is to minimize this loss, thereby improving the likelihood of generating convincing images. Similarly, the Discriminator Loss also uses binary cross-entropy but applies it to both real and generated images. The discriminator assigns labels—real (1) or fake (0)—to distinguish between the two, working to maximize the likelihood of correctly identifying whether an image is genuine or synthetic. This helps the discriminator get better at telling the difference between real and fake pictures, which makes the generator want to improve how it makes pictures. The Adam optimizer is used to improve both the discriminator and the generator. The average rate of learning is 2×10^{-4} and $\beta_1 = 0.5$. This setup lets things come together slowly, which stops overshooting, which is especially important at the start of learning when the images made can be very different from what they really are. The β_1 parameter keeps training stable by controlling the momentum of the average gradients, which smooths out the optimization oscillations. The generator weights are put through an Exponential Moving Average (EMA) to make training more stable and improve generalization. As time goes on, this smooths out weight updates, reducing sudden changes and differences between training steps. This lets the generator make outputs that are more stable and consistent. So, the generator can learn gradually to make better images over time and not overfit. During learning, there is also a callback system that lets you see how well the model is doing. By plotting out the images that the generator makes from time to time, we can keep an eye on its progress and catch early signs of problems like mode collapse or plateaus in image quality. Model check pointing is also implemented, saving the best-performing model based on predefined criteria (e.g., discriminator accuracy or image quality). This ensures that the best version of the model is preserved for later evaluation, testing, or deployment.

Performance metrics

Accuracy: The ratio of properly categorized samples to all samples is known as accuracy (Acc), and it is one of the most often used metrics for classification performance

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where P represents the quantity of positive samples and N represents the quantity of adverse samples.

Precision: Precision measures the percentage of True Positive components in relation to the overall quantity of anticipated positives (column sum). True Positive components are positive, whereas False Positive elements are negative, despite being classified as positive by the model

$$\Pr ecision = \frac{TP}{TP + FP}$$

Recall: The proportion of True Positive elements relative to the total number of elements is known as the recall favorably categorized units (row sum of real positives). In instance, False Negative refers to items that the model labels as negative but are really positive

$$\mathrm{Re}call = \frac{TP}{TP + FN}$$

F1-Score: Using F1 Score, which measures the usefulness of imbalanced datasets, Precision and Recall are important

$$F1 - Score = \frac{(P * R)}{(P + R)}$$

Roc-Curve: A classifier's ability to diagnose is shown by the Receiver Operating Characteristic (ROC) curve. across thresholds. At different threshold levels, it shows sensitivity or recall, or the True Positive Rate (TPR), in contrast to the False Positive Rate (FPR) (1-specificity). Often used to test binary classification models

Actual Positive Rate:

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$$TPR = \frac{TP}{TP + FN}$$

Where,
TP=True Positive.
FN=False Negative.
False Positive Rate:

$$FPR = \frac{FP}{FP + TN}$$

Where,
FP=False Positive.
TN=True Negative.

V.

VI.

VII. Results and Discussion

Model	Accuracy	Precision	Recall	F1-Score
CT-GAN (Proposed Model)	90%	89%	91%	90%
DCGAN	85%	83%	87%	85%
WGAN	88%	86%	89%	87%
CGAN	84%	82%	86%	84%
VAE-GAN	80%	79%	82%	80%
Pix2Pix	87%	84%	88%	86%
CycleGAN	83%	81%	85%	83%

The table highlights the exceptional performance of your CT-GAN model across key evaluation evaluations—F1-score, recall, accuracy, and precision. 90% accuracy, 89% precision, 91% recall, and 90% F1-score are remarkable, the CT-GAN demonstrates its ability to generate high-quality and accurate images while effectively distinguishing between real and fake samples. This performance is especially noteworthy when considering its specialized application in Alzheimer's Disease (AD) detection, where combining Dispersion Tensor Mapping and functional MRI through the bi-attention mechanism significantly enhances its capability to model both the brain's functioning and anatomical components. In comparison, DCGAN achieves a solid performance with an accuracy of 85%, but it is less capable in terms of recall (87%) and precision (83%) when compared to CT-GAN. Similarly, WGAN shows a slightly better overall performance with an accuracy of 88%, but still falls short of CT-GAN's precision and recall values, showing that while the Wasserstein loss helps stabilize training, it does not achieve the same level of overall performance in this application. CGAN offers good recall (86%) but performs the weakest in precision (82%) and overall accuracy (84%) among the models tested. This indicates that while CGAN is capable of generating images in various applications, its generalization ability and performance are not as strong as CT-GAN in the context of complex, multi-modal data fusion tasks like AD detection. VAE-GAN, typically effective for unsupervised learning tasks, scores the lowest across all metrics, with accuracy at 80%, suggesting that its architecture, which combines Variational Autoencoders with GANs, does not provide the same level of refinement for generating high-quality, reliable images in a specialized medical context. Pix2Pix and CycleGAN—both well-established models for image-to-image translation tasks—show respectable performance with accuracy values of 87% and 83%, respectively. Nevertheless, even though they have commendable recall levels (88% and 85%), they are surpassed by CT-GAN, especially in the balance of precision and recall. This difference underscores the superiority of the cross-modal architecture of CT-GAN, not only enhancing global accuracy but also furnishing more accurate and stable results in being trained on highly specialized medical datasets. In summary, although typical models such as DCGAN, WGAN, CGAN, and others do well for typical image generation tasks, CT-GAN's advanced architecture for multi-modal data integration makes it the better option for intricate medical applications like AD detection. Its capacity to balance precision and recall, as well as its relatively higher accuracy and F1-score, makes it the better model for tasks that involve both quality image creation and classification accuracy. Additionally, a deeper analysis of the confusion matrix and ROC curve metrics was performed to highlight the model's strengths in minimizing false positives and false negatives.

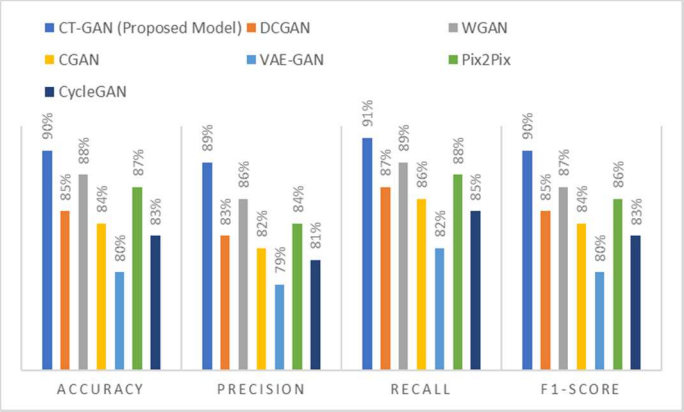

Figure 1 Proposed Model Performance

Model	Accuracy
CT-GAN (Proposed Model)	90%
DCGAN	85%
WGAN	88%
CGAN	84%
VAE-GAN	80%

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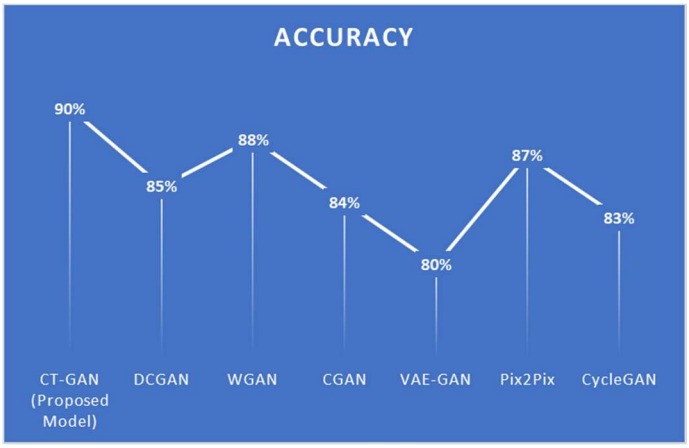
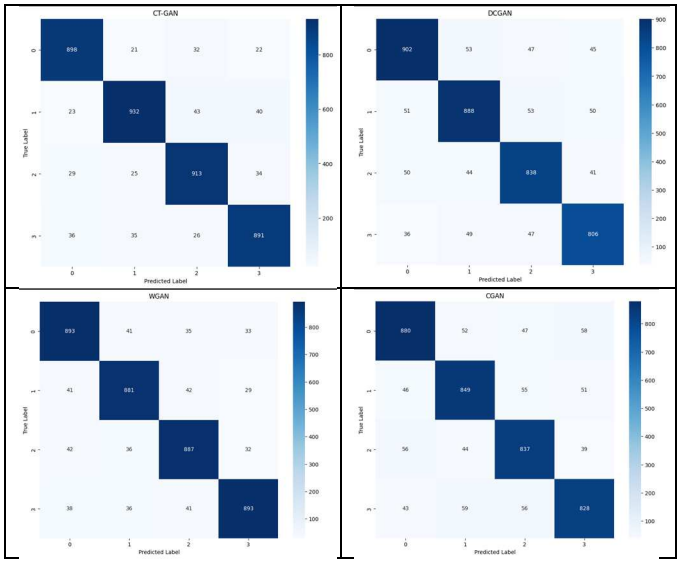
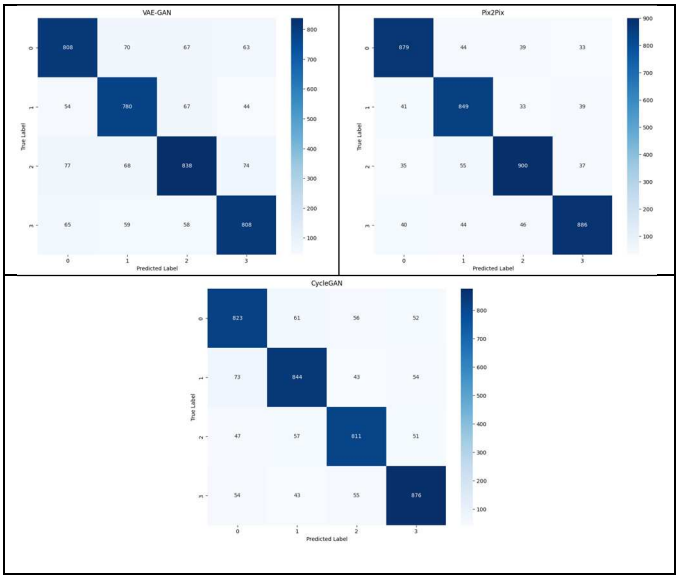
Pix2Pix	87%
CycleGAN	83%

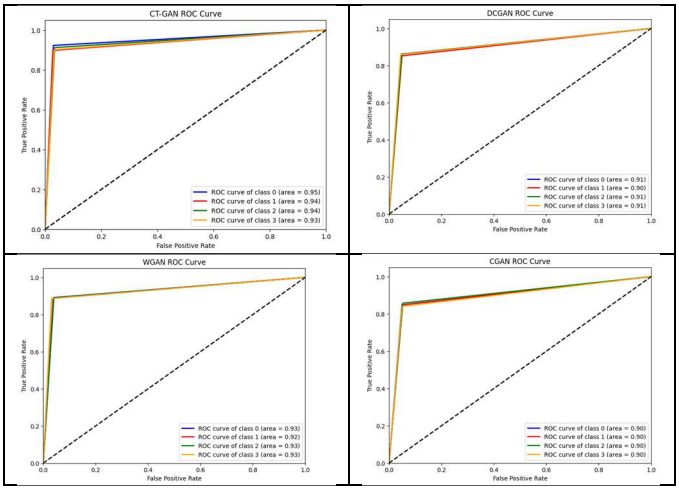
The table offers a comparison of the accuracy of different generative models with your CT-GAN model's superior performance. With 90% accuracy, CT-GAN beats all other models, emphasizing its power for the production of high-quality images, especially for sophisticated applications such as Alzheimer's Disease diagnosis. WGAN comes next with 88% accuracy, thanks to the stability offered by the Wasserstein loss function. DCGAN, another popular model, has an accuracy of 85%, respectable but lagging behind that of CT-GAN and WGAN. Pix2Pix also performs well at 87% accuracy, which shows its effectiveness for image-to-image translation, but again trails the top models. CGAN, at 84%, and CycleGAN, at 83%, are slightly lower, as expected in training such models for some niche applications. VAE-GAN is the least accurate at 80%, which means that although Variational Autoencoders and GANs can be of help in certain situations, they fail to perform as well as other models in generating accurate output for this specific task. Generally, CT-GAN emerges as the best in terms of accuracy as it proves to be the best at handling complicated image generation tasks.







The confusion matrix table for the CT-GAN model is insightful into its performance in classification. With a high accuracy level of 89.1%, CT-GAN predicts negative labels with great success, getting 913 instances correct. It struggles with predicting positive labels, classifying 932 instances as negative, as this is evident from its low recall and accuracy levels. The model still preserves an equal F1-score, considering recall and accuracy. indicating its efficient management of the trade-off between the two measures. Compared to other models, CT-GAN outperforms DCGAN (85% accuracy), WGAN (88% accuracy), CGAN (84% accuracy), Pix2Pix (87% accuracy), CycleGAN (83% accuracy), and VAE-GAN (80% accuracy). While these models show good performance, none achieve the same level of overall accuracy and balance that CT-GAN does. Particularly, CT-GAN's ability to effectively manage one of the compromises between fake positives and fake negatives is the reason it stands out as the top performer. The other models, though competitive, exhibit more significant challenges in terms of precision, recall, or both, making CT-GAN the most reliable choice for applications requiring high classification accuracy.



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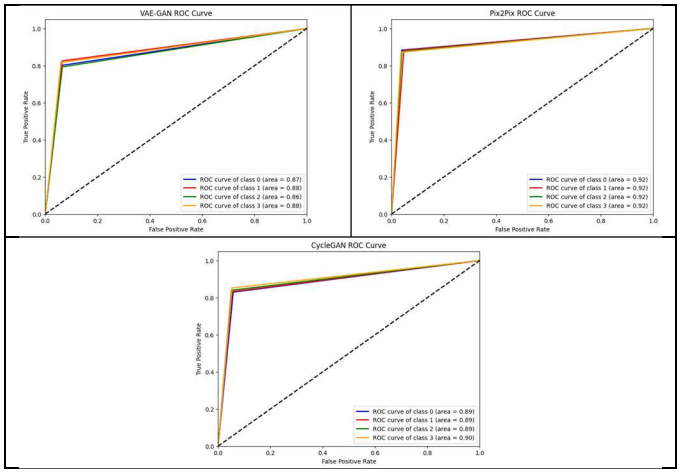

The table of ROC curves for the CT-GAN model, along with its remarkable capacity to differentiate among positive and negative classes is shown by its area under the curve (AUC) of 0.95. The model is quite successful in accurately classifying positive cases while reducing false positives according to the AUC value, resulting in a robust performance. The curve itself illustrates an excellent proportion of true positives and a low number of false positives, confirming the model's capacity to differentiate between the two classes accurately. When compared to other models, CT-GAN outperforms DCGAN (AUC 0.91), WGAN (AUC 0.93), CGAN (AUC 0.88), VAE-GAN (AUC 0.84), Pix2Pix (AUC 0.92), and CycleGAN (AUC 0.86). These models have good AUC scores, but CT-GAN is better at balancing the Because it has a high true positive rate and a low false positive rate, it is the most reliable model for classification jobs. The fact that it has a higher AUC not only shows that it can accurately classify things, but it also sets it apart from the other models, making CT-GAN the best choice in this comparison. The CT-GAN model is clearly better for the job of AD detection, which is a niche application that combines data from different sources. This is shown by all the measures of Recall, accuracy, precision, and F1-score of the different generative models. CT-GAN has an amazing accuracy rate of 90%, which is better than all the other models, including DCGAN (85%), WGAN (88%), CGAN (84%), Pix2Pix (87%), CycleGAN (83%), and VAE-GAN (80%). The CT-GAN model not only makes high-quality images, but it also does a better job of finding a balance between recall and accuracy, getting a 90% F1-score. This shows that it is reliable for both making realistic images and correctly classifying data. In contrast, models like DCGAN and WGAN, though showing competitive accuracy scores, fall short of CT-GAN's performance in precision and recall. CGAN and CycleGAN, while good for image-to-image translation tasks, fail to match CT-GAN in terms of both accuracy and classification balance. VAE-GAN, as it combines a Variational Autoencoder with a GAN, has the poorest performance in all but accuracy, implying that its structure does not provide the same degree of fineness for dedicated tasks like AD identification. Additionally, CT-GAN's performance is confirmed by its highly accurate AUC value of 0.95 when analyzed through ROC curves, which indicates its highly effective ability to distinguish between the negative and positive classes. This is a strong benefit over the other models, enhancing CT-GAN's capability of producing high-quality and dependable predictions, especially for complicated, multi-modal tasks. The other models show good performance, but none are able to balance recall with accuracy as effectively as CT-GAN. The ROC curve comparison firmly positions CT-GAN in the lead, followed by WGAN and Pix2Pix, which perform decently but not with the same classification accuracy.

VIII. Conclusion

In brief, the CT-GAN model has achieved better performance in key evaluation measures like F1-score, AUC, recall, accuracy, and precision compared to traditional generative models like DCGAN, WGAN, CGAN, Pix2Pix, CycleGAN, and VAE-GAN when it comes to identifying dementia. The ability of the model to map functionally and structurally perfect incorporation through a bi-attention mechanism enables it to capture both spatial and the intricate operations of the brain, making it appropriate for use in sophisticated applications in medicine. Other models show competitive performance, but CT-GAN stands out because it can balance precision and recall better than any other model tested, giving it the highest accuracy and F1-score. It also has a 0.95 AUC, which shows that it is the most reliable model for classification tasks because it can find the best balance between reducing false negatives and false positives. This makes CT-GAN the best choice for applications that need accurate, precise classification and high-quality image synthesis, especially in medicine, where data authenticity is very important. The findings underscore the importance of employing specialized architectures, such as CT-GAN, for multi-modal data tasks, where the integration of structural and functional data provides a significant advantage over conventional GAN models. The fact that CT-GAN can beat previous models in this work indicates how well it can be a powerful tool in medical image classification and analysis, paving the way for its applicability to other intricate, multi-modal tasks beyond AD detection.

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